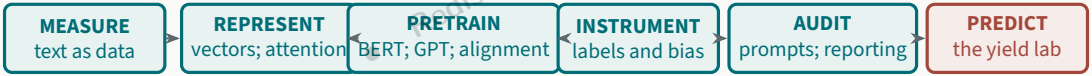
Session 5.6

Text Signals and the Bond-Yield Forecast

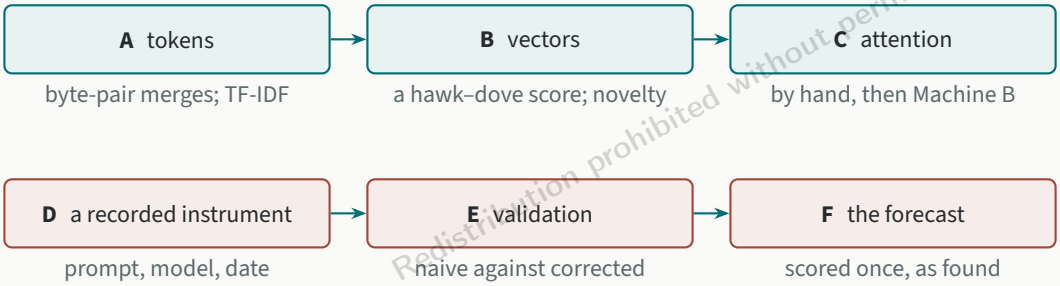
Four signals, one frozen baseline, one scoring of the test window, and the result as it came out



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Zhigang Feng

Six Blocks, and What Each One Has to Clear



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- Blocks A to D build four signals out of one corpus; E asks what a validation set buys; F scores Session 2's forecast once.
- **Block F's design was hashed before its code existed. No gate depends on text helping.**

Source: The lab, Session 5, blocks A to F; the pre-registered protocol for block F, hashed 2026-09-20.

The Rules of the Lab

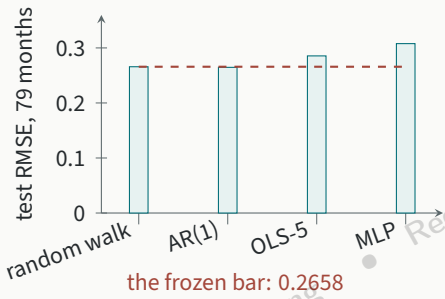
Block	Runs live	Recorded, and quoted on these slides
A	200 merges; TF-IDF on 149 statements	the same, and the segmentation table
B	co-occurrence, PPMI, 100 dimensions	the same, and the full validation table
C	attention by hand; 60 training steps	Machine B: maps, curves, three temperatures
D	reads a recorded score file	245 statements labelled, 48 four ways
E	the chapter's own scripts	the same, and a sweep over validation sizes
F	the validation window	the test window, scored once against the file

- Offline, on a CPU, from a clean kernel, fixed seeds. Imports: numpy, pandas, torch.
- **Every number on these slides is in a recorded file the notebook prints beside its own.**

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Source: The lab, Session 5; recorded run of 2026-09-20.

Session 2's Frozen Baseline, Restated in Full

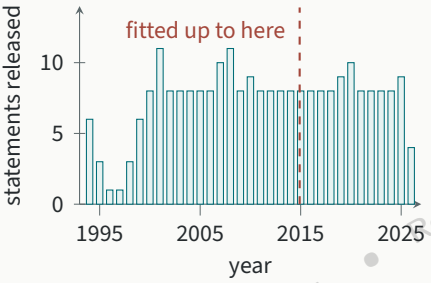


- One month ahead, the ten-year yield, 79 months. Session 2's three reproducible rows must return to four decimals, or the run stops.
- The MLP row is two objects: the published 0.3078 averages ten per-seed errors; the mean forecast scores 0.2956.
- **The bar is the random walk, 0.2658 — AR(1) all but ties it — and it is a very low-variance forecast.**

Source: Session 2's frozen baseline file, scored over 79 months; Deck 2.4.

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The Corpus, and the Window Everything Is Fitted On



- 245 statements, 1994-02-04 to 2026-06-17; 149 fall in the training window, which ends 2014-12-31.
- Every fitted object is fitted there; 2015–2019 chooses penalties; the test window enters no choice.
- **A month's signal is the last kept statement dated on or before that month's last observation, and no later document reaches the row.**

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Source: The lab's frozen corpus manifest; the pre-registered protocol, information rule.

Five Releases That Are Not Policy Decisions

Date	What it is	Why it is not a policy decision
2007-08-10	a market-functioning notice	the Desk supplies reserves; the target is unchanged
2008-03-11	a facility announcement	a lending facility and swap lines, with four other banks
2010-05-09	a swap-line renewal	a balance-sheet action, no rate decision, no stance
2019-10-11	a reserve-management notice	bill purchases and repos, called purely technical
2025-08-22	a framework review	revisions to the longer-run goals; no view on the stance

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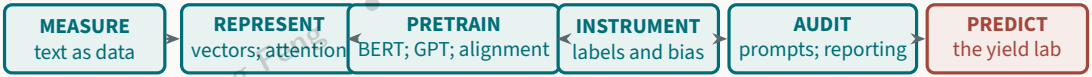
- 7 months carry two or three releases; the last kept one wins, and 240 remain.
- **The list was read, not sampled, and it sits inside the hashed protocol.**

Source: The pre-registered protocol, exclusion set; the Committee's own release texts.

Session 5.6

Four Signals Out of One Corpus

Blocks A to D: tokens, vectors, attention, and an instrument that reads



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Block A: Byte-Pair Merges, Learned by Hand

the first merges, in order

`e + _` `i + n` `t + h` `th + e_`

a word the window uses often

raising
ra | is | ing_

a word it never uses

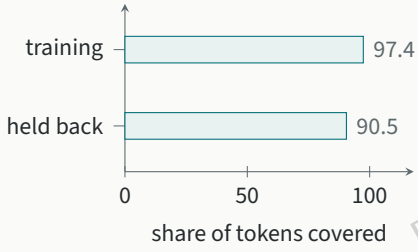
reflationary
re | f | l | a | tion | ar | y_

- 200 merges on the 149 training statements: count adjacent pairs, merge the commonest, repeat.
- The merges follow this corpus's frequencies, not English's.
- "Raising" takes 3 pieces; "reflationary" is not in the 1,628-word vocabulary and takes 7.
- **Change the window and the same word splits differently.**

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Source: Sennrich, Haddow & Birch (2016); Miao & Feng, *AI for Economists*, ch. 13; the lab, block A.

Block A: TF-IDF, Fitted Where It Is Allowed To Be

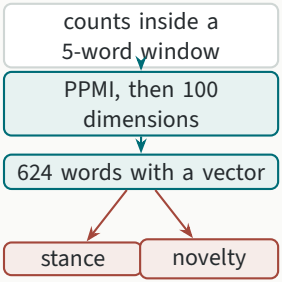


- A term is kept if it appears in at least 2 training statements: 1,163 columns, and 85.5 per cent of the 245-row matrix is zero.
- Coverage falls from 97.4 per cent of training tokens to 90.5 per cent of held-back tokens: the vocabulary ages.
- **Fitting on all 245 statements would raise the second bar and destroy the experiment.**

Source: The lab, block A, TF-IDF fitted on the 149 statements of the training window.

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Block B: A Hawk–Dove Axis and a Novelty Index

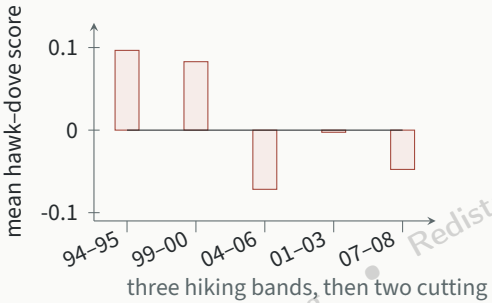


- Count which words occur near which, weight by pointwise mutual information, truncate to 100 dimensions.
- **Stance** is closeness to 5 hawkish seeds minus closeness to 4 dovish ones. **Novelty** is one minus the cosine between consecutive statements, and is zero in a month with no meeting.
- **Both are fitted inside the training window — the property the next page puts on trial.**

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Source: The lab, block B; Miao & Feng, *AI for Economists*, ch. 13, on distributional representations.

Block B: The Embedding Score Fails Its Validation



- The outside standard is what the Committee did next: hawkish scores on bands of increases, dovish on cuts.
- The 2004–06 hiking cycle, 17 statements, scores -0.072 — the most dovish stretch in the sample — while two other hiking bands score 0.097 and 0.083.
- **Rank correlation with the next 3 meetings' path is $+0.07$. That is the result, not a bug to be repaired.**

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Source: The lab, block B, validated on the training window against the subsequent policy path.

Block B: The Rule That Passes, and What It Actually Is

signal	hiking bands	cutting bands	rank corr.	verdict
embedding hawk–dove	0.036	-0.025	$+0.07$	fails
decision sentence	0.810	-0.568	$+0.68$	passes
novelty index	0.189	0.180	-0.06	fails

- The rule that passes is two regular expressions — 7 tightening verbs, 9 easing verbs — read on the sentence that says what the Committee decided.
- **That sentence restates the funds-rate change the panel already carries. A signal can pass validation by re-deriving a number you have.**

Source: The lab, block B, three signals scored on 149 statements of the training window.

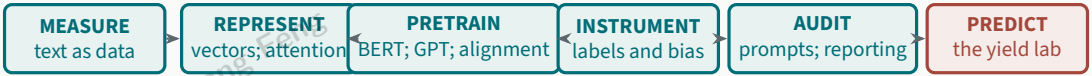
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Session 5.6

Inside the Model, and Beside It

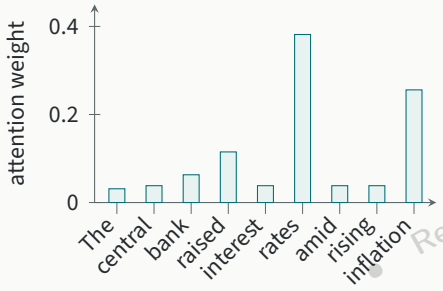
Block C: attention by hand, then Machine B. Block D: a score whose information set you cannot date

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Block C: Attention by Hand, Against the Book



- The chapter stipulates 9 scores for one sentence, with “interest” as the query, and prints the weights.
- The lab recomputes the softmax: the exponentials sum to 26.13, “rates” takes 0.382 and “inflation” 0.256, and all 8 printed figures agree.
- **Doing the arithmetic is how you learn the example was stipulated, not produced by a model.**

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Source: Miao & Feng, *AI for Economists*, ch. 13, worked attention example; the lab, block C.

Block C: Machine B's Attention, Layer by Layer

	head 1	head 2	head 3	head 4
layer 1	3.27	3.25	3.30	3.10
layer 2	2.04	1.68	1.00	1.74
layer 3	1.49	1.57	1.11	1.52

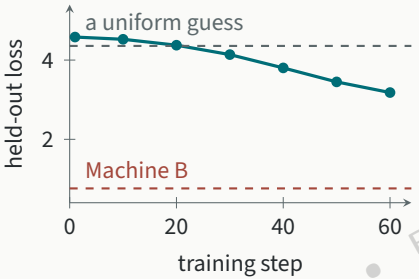
mean row entropy on one sentence; 4.32 is the flattest possible

- **Machine B** is Session 1's character-level transformer: 631,424 parameters, 3 layers, 4 heads, trained 3,000 steps to a held-out loss of 0.761.
- Layer 1 stays diffuse in all four heads; by layer 2 one head has collapsed to 1.00.
- **Nobody asked for a division of labour between heads; training produced it.**

Source: The lab, block C, on Session 1's frozen model and a 75-character sentence.

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Block C: Sixty Steps, and Three Temperatures

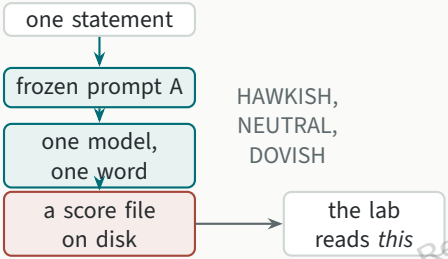


- Live: 60 steps, about 15 seconds. Loss 4.579 to 3.260, where a uniform guess over 78 characters scores 4.357.
- Machine B's 3,000 steps take about 18 minutes. Sampled at 0.5, 0.8 and 1.2, its phrasing holds at the lowest and dissolves at the highest.
- **Fluent phrasing is the cheap part of the objective, not evidence about policy.**

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Source: The lab, block C; the live run and Session 1's recorded training curve.

Block D: A Stance Score as a Recorded Instrument



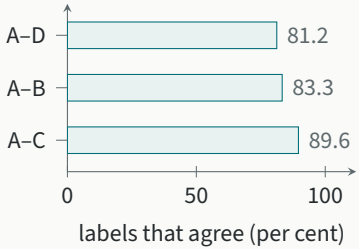
- 245 statements, one frozen prompt, one word of output mapped to $+1, 0, -1$: 84 hawkish, 61 neutral, 100 dovish.
- Every reply is kept with its date, the prompt's hash and the model that answered. The lab never calls a model; it reads a file.
- **A number a model produced is reproducible only if the run was recorded.**

claude-sonnet-5 through a 2.1.278 harness · 2026-09-20 · prompt A, 846fe144b241d9d1

Source: The lab, block D; the recorded stance run of 2026-09-20, 389 calls in all.

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Block D: The Same Prompt, Twice

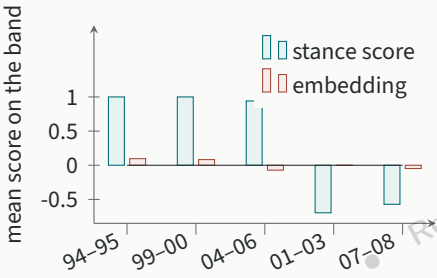


- Four variants on the same 48 statements. B rewords the instruction, C reverses the order of the labels, D is A again, character for character.
- A and D agree on 81.2 per cent: 9 statements change their label when nothing changes. Rewording gives 83.3, reordering 89.6.
- **Prompt sensitivity cannot be read off a gap smaller than the instrument's own noise.**

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Source: The lab, block D, four prompt variants on a 48-statement subsample.

Block D: The Policy Path as an Outside Check

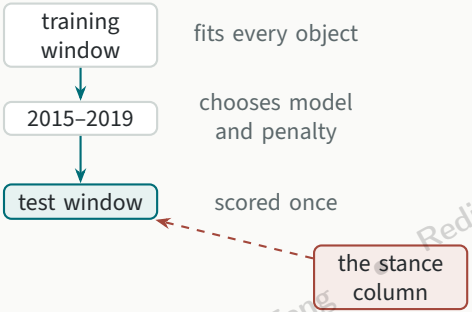


- Same bands, same training window: hiking 0.980 against cutting -0.634, and rank correlation $+0.49$ with the subsequent path.
- On the 2004–06 cycle the embedding read backwards, the stance score reads 0.941 — with no seed list and no fitting.
- **It is also the one signal whose information set nobody here can date. Hold that until the last pages.**

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Source: The lab, blocks B and D, scored on the same 149 statements of the training window.

Block D: Reported Beside the Forecast, Never Inside It



- Only variant A, run on every statement, may enter a forecast; B, C and D are diagnostics.
- No model choice and no penalty may depend on the stance column: it is scored beside the structured columns, never pooled into them.
- **Machine B is excluded by assertion — it was trained on all 245 statements, the test window included.**

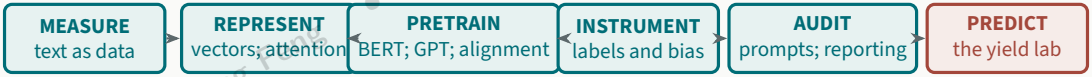
Source: The pre-registered protocol, sections on the language-model column and on excluded artefacts.

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Session 5.6

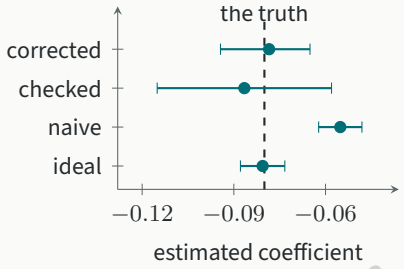
What Validation Buys, and the Forecast

Blocks E and F: the book's own simulations, then one scoring of the test window



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Block E: Naive, Checked-Only, Corrected

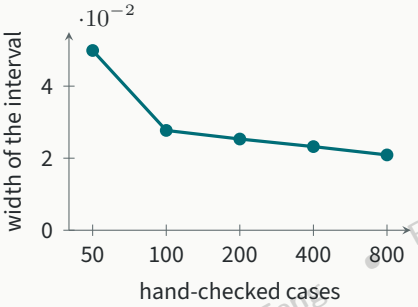


- The chapter's simulation: 3,000 observations, a true coefficient of -0.080, a label correlated 0.851 with the truth, 200 checked by hand.
- Treating the label as the truth returns -0.055, 31.0 per cent of the way to zero, with an interval that excludes it.
- **Calibrating on the checked cases is unbiased and 48.7 per cent narrower than using them alone.**

Source: Miao & Feng, *AI for Economists*, ch. 14, bias-correction simulation, run unchanged.

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Block E: What a Validation Set Buys

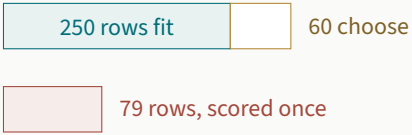


- The interval narrows 49 per cent between fifty hand-checked cases and 200, and a further 17 per cent between 200 and eight hundred.
- What those cases estimate is the calibration slope: 0.611 at fifty against 0.753 at 200.
- **Most of what a validation set buys, it buys early. Budget for a few hundred.**

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Source: Miao & Feng, *AI for Economists*, ch. 14, bias-correction script; the lab, block E.

Block F: Scored Once, and the Power to Detect a Gain Is Low

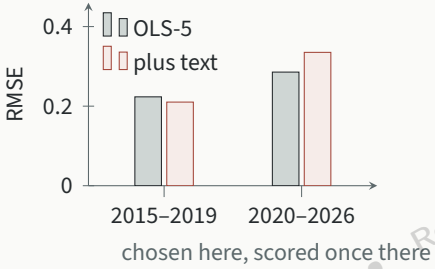


- The sample starts at 1994-02-28: no earlier row could carry a text signal. 250 rows fit, 60 choose, 79 are scored once, after every choice was made.
- All 7 gates are on procedure: hashes, the frozen baselines, the split sizes, the inherited gap, no post-cutoff document in a fitted object.
- **Say this before the table: with 79 months the power to detect a small gain is low. Not significant is not the same as useless.**

Source: The pre-registered protocol, sample definition and gates; hash 42ae9f04550d6f29, taken 2026-09-20.

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Block F: On the Window Where It Was Chosen, Text Helped



- The model and the ridge penalty were chosen on 2015–2019, and nowhere else. There, text *helps*: 0.2234 against 0.2102, or 5.9 per cent better.
- On the test window the same pair is 0.2856 against 0.3352: 17.4 per cent worse.
- **A signal can look good exactly where you chose it. That is what a hashed protocol is for.**

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Source: The lab, block F: validation metrics recomputed live, test metrics from the recorded file.

Block F: The Test Window, Scored Once

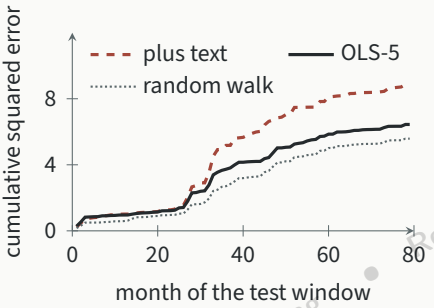
	frozen file		re-estimated		plus text	
model	RMSE	MAE	RMSE	MAE	RMSE	MAE
random walk	0.2658	0.2097	0.2658	0.2097	–	–
AR(1)	0.2647	0.2089	0.2648	0.2090	–	–
OLS-5	0.2855	0.2255	0.2856	0.2247	0.3352	0.2567
OLS-5 + TF-IDF ridge	0.2855	0.2255	0.2856	0.2247	0.3364	0.2584
text alone	–	–	–	–	0.2685	0.2105
TF-IDF ridge alone	–	–	–	–	0.2675	0.2099
LLM stance column	–	–	–	–	0.2714	0.2146

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- Re-estimating costs 0.2855 to 0.2856; the third column is the text's doing.
- **Text does not help: added to OLS-5 it is clearly worse, and alone it lands behind the frozen bar, 0.2658.**

Source: The lab, block F, one scoring of 79 test months; Session 2's file in the first pair.

Block F: The Path, and Two Statistics That Agree



- Diebold–Mariano, Harvey–Leybourne–Newbold corrected: -2.817 and -2.948 ($p = 0.006, 0.004$), both favouring plain OLS-5.
- Clark–West, built to find such a gain, gives -0.597 and -1.777 and declines to.
- Text alone against the frozen bar: -1.165, $p = 0.248$. Indistinguishable.
- **The text model leads for 6 months of 79 and trails for 73.**

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Source: Diebold & Mariano (1995); Harvey, Leybourne & Newbold (1997); Clark & West (2007); the lab, block F.

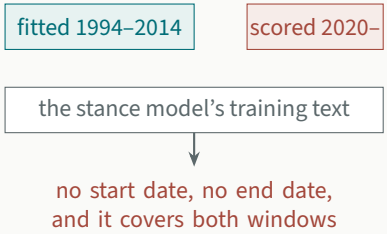
Where This Can Mislead: Chosen on One Window, Scored on Another

- The finding is not “text does nothing”. Text looked like it helped on the window where the model was chosen, by 5.9 per cent, and then cost 17.4 per cent on the window scored once.
- A student who sees only the test column learns that text is useless. One who sees both learns why the protocol file has a hash, and why the hash was taken before the code ran.
- Four signals, two scopes for the decision rule, several validations and 5 penalties were tried — validation pushed the ridge to 100, shrinking the text almost away, and it still landed behind the bar.
- Report the arms that failed, or what survives is a selected maximum with fictitious error bars.
- **The gap between the validation column and the test column is the size of the mistake you would have published.**

Source: The lab, block F, validation and test windows; the pre-registered protocol, gates and reporting rule.

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Where This Can Mislead: An Instrument You Cannot Date



- What the stance model knows about a 2022 statement includes what was written after it, and where the yield went.
- 55 statements span the 79 test months, 26 of them repeats. The panel is today's vintage; the text is the only real-time part.
- **The clean signals are those trained inside the training window; the most capable instrument is the one you cannot date.**

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Source: The pre-registered protocol, stale-month rule and excluded artefacts; the lab's corpus manifest.

Your Turn, and Where This Corpus Goes Next

- Pick one. Refit the embedding on all 245 statements and watch the 2004–06 band. Narrow the decision-sentence rule and rerun its validation. Add the stance column to OLS-5 and report both tests.
- Report the change you made, the validation table before and after, and one sentence on what moved. If a signal passes, say what it correlates with that you already had.
- What block F answers: on this window, against this bar, text does not improve the forecast, and the two statistics agree. What it does not answer: anything causal, and anything about a window with more meetings in it.
- **Session 6 takes these same 245 statements as a library to retrieve from, and the question becomes whether it found the right passage and whether you can tell.**

Source: The lab, Session 5, blocks A to F; the corpus carried into Session 6's retrieval lab.

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